# XJTLU Entrepreneur College (Taicang) Cover Sheet

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| Module code and Title | DTS402TC - Machine Learning | |
| School Title | School of Artificial Intelligence and Advanced Computing | |
| Assignment Title | Group Coursework | |
| Submission Deadline | 5 pm China time (UTC+8 Beijing) on Sunday 28th Dec 2025 | |
| Final Word Count | NA | |
| If you agree to let the university use your work anonymously for teaching and learning purposes, please type **"yes"** here. | | |

I certify that I have read and understood the University's Policy for dealing with Plagiarism, Collusion and the Fabrication of Data (available on Learning Mall Online). With reference to this policy I certify that:

- My work does not contain any instances of plagiarism and/or collusion.
  My work does not contain any fabricated data.

**By uploading my assignment onto Learning Mall Online, I formally declare that all of the above information is true to the best of my knowledge and belief.**

| **Scoring – For Tutor Use** | |
|---|---|
| **Student ID** | |

| **Stage of Marking** | | | **Marker Code** | **Learning Outcomes Achieved (F/P/M/D) (please modify as appropriate)** | | | **Final Score** |
|---|---|---|---|---|---|---|---|
| | | | | **A** | **B** | **C** | |
| 1st Marker – red pen | | | | | | | |
| Moderation – green pen | | | **IM Initials** | The original mark has been accepted by the moderator (please circle as appropriate): | | | Y / N |
| | | | | Data entry and score calculation have been checked by another tutor (please circle): | | | Y |
| 2nd Marker if needed – green pen | | | | | | | |
| **For Academic Office Use** | | | | **Possible Academic Infringement (please tick as appropriate)** | | | |
| **Date Received** | **Days late** | **Late Penalty** | | ☐ **Category A** | Total Academic Infringement Penalty (A,B, C, D, E, Please modify where necessary) ______________ | | |
| | | | | ☐ **Category B** | | | |
| | | | | ☐ **Category C** | | | |
| | | | | ☐ **Category D** | | | |
| | | | | ☐ **Category E** | | | |

**DTS402TC Machine Learning**

# Group Coursework

**Due:** Sunday Dec. 28th, 2025 @ 17:00
**Weight:** 50%
**Maximum score:** 100 points (70 for group work + 30 for individual work)

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## Learning Outcomes Assessed

- C. Systematically evaluate the performance of machine learning models using appropriate metrics and techniques.
- D. Analyse machine learning experimental results, interpret findings, and critically appraise outcomes and conclusions.
- E. Use appropriate techniques to devise the best implementation of machine learning algorithms to solve practical learning problems.

## Team Policy

You are free to form your group up to **3** team members (with a minimum of two members), you need to submit all team members' information before 13th Nov, 23:59 via LMO. Students who fail to do so will be randomly assigned to a team. Changes will not be allowed once the teams have been confirmed.

## Coding Policy

- **Programming Language:** All code must be written in Python, with a version of 3.6 or higher. Students are required to use ***Jupyter Notebook*** for their implementation, and the main file must be an '`.ipynb`' file that includes all outputs (such as printed results, graphs, etc).
- **Code Structure:** The code must be well-structured, readable, and properly documented. Comments should be included for each function and section, explaining its purpose and functionality.
- **Libraries and Tools:** Usage of high-level machine learning libraries (e.g., scikit-learn, TensorFlow, PyTorch, Keras, XGBoost, etc.) for model construction is prohibited. However, basic libraries such as NumPy, Pandas, and Matplotlib are allowed for data handling, manipulation, and visualisation.
- **Code Validation:** Ensure the code is fully executable without errors. All notebooks must be submitted with outputs included (e.g., model training results, visualisations, evaluations), allowing the marker to verify the results without rerunning the notebook.

## Submission Policy

1. **Submission Format**
   - Each team must submit a single zip file containing:
     (a) The final report in **PDF** format.
     (b) A folder labelled `code` that contains all individual model implementations, data preprocessing scripts, and evaluation scripts. Each individual's work should be in a separate Jupyter Notebook (`.ipynb`) file.
     (c) A `README` file explaining how to run the code, specifying any dependencies (libraries) and installation instructions.
2. **File Naming**
   - The file and folders should be named in the format: `TeamName_Coursework.zip`. Each individual's code must be in a separate folder, e.g., `TeamName_JohnDoe_StudentID_Model.zip`.

## Late Policy

5% of the total marks available for the assessment shall be deducted from the assessment mark for each working day after the submission date, up to a maximum of five working days.

## Academic Integrity and Collaboration Rules

- You may **discuss ideas and concepts** with your team members.
- Your **individual model implementation and code** must be completed **independently**.
- Do **not share, reuse, or modify** other members' code for the individual component.
- Do **not share or read** code/work from other teams; discussions between teams should remain **high-level only**.
- Do **not submit** work from other teams.
- The use of **AI tools** (e.g., *ChatGPT, Copilot, etc.*) to **generate or complete** any part of the work is **strictly prohibited**.

# 1 Problem Selection & Data Collection (20%)

1. **Problem Selection (5%)**
   - Choose a practical real-world problem (e.g., image classification, text classification, time-series forecasting, anomaly detection, or multimodal learning)
   - Justify the problem choice by explaining its significance and the challenges involved.
2. **Dataset Selection (5%)**
   - Select an open-source dataset with **at least 20 dimensions** and **over 2000 data points**.
   - Avoid using simple open-source datasets such as MNIST, CIFAR-10, CIFAR-100, Iris Flower, or those used in lab sections.
   - If using an open-source dataset, specify the **download address**.
   - Collect at least 1000 samples of your own data with the same schema/features as the open-source dataset. Synthetic or artificially generated data (e.g., using NumPy random values) is not allowed.
   - Provide details on data source, collection procedure, and quality assurance.
3. **Dataset Comparison (5%)**
   - Compare open-source and collected datasets in terms of:
     - Feature distributions (statistical summaries, visualizations).
     - Class balance, and missing values.
     - Potential biases or representativeness.
4. **Data Preprocessing (5%)**
   - Document and justify preprocessing for both datasets:
     - Cleaning, normalization, encoding, missing value handling.
     - Domain alignment (how to make the collected dataset comparable to the open-source one).

# 2 Model Selection & Implementation (*Individual Section*, 30%)

*This section will be scored individually. Each team member must clearly indicate in their report and code which model they implemented.* ***If team members fail to clearly indicate their individual contribution, the lowest individual score from the team will be given to all team members.***

1. **Model Selection & Implementation (15%)**
   - Each team member must select and implement a different type of machine learning model. To ensure methodological diversity, models should be chosen from distinct categories. Examples include:
     - **Linear and Kernel-based Models**: Logistic Regression with regularization, Support Vector Machines (linear or kernel-based).
     - **Tree-based and Ensemble Methods**: Random Forest, XGBoost, LightGBM, etc.
     - **Neural Networks**: Multilayer Perceptron (MLP), Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), Transformers.

  - **Probabilistic and Graphical Models**: Naïve Bayes, Gaussian Mixture Models, Hidden Markov Models.
- For unstructured data such as images or text, students may apply established **feature extractors** before model training, such as:
  - **Image Features**: SIFT, HOG, or pretrained CNN feature embeddings (e.g., ResNet, VGG).
  - **Text Features**: Bag-of-Words, TF-IDF, or pretrained embeddings (e.g., Word2Vec, GloVe, BERT).
- Justify why the chosen model is appropriate for the task.

2. **Model Training and Reporting (15%)**
   - Models must be trained on a **mixed dataset** (open-source + collected data).
   - Appropriate **hyperparameter tuning** must be conducted and reported.
   - Training and validation strategies (e.g., cross-validation, hold-out sets) must be clearly described.
   - Each student must submit a separate Jupyter Notebook documenting their full pipeline, including preprocessing, model training, results, and code outputs.
   - The report must include clear tables, figures, and textual explanations that highlight the contribution of each individual model.

## 3 Result Analysis and Evaluation (30%)

1. **Evaluation Metrics (10%)**
   - The team must select and apply appropriate evaluation metrics relevant to the chosen task (e.g., accuracy, F1-score, ROC-AUC for classification; RMSE, MAE for regression).
   - Evaluate the models using a variety of criteria, such as predictive accuracy, computational efficiency, model complexity, and memory consumption.
2. **Comparative Analysis (10%)**
   - Conduct a comprehensive comparison of all implemented models using the selected evaluation metrics.
   - Perform **cross-dataset analysis**: evaluate performance separately on open-source data, collected data, and mixed datasets.
   - Analyse trade-offs between models, considering predictive performance, generalization capability, interpretability, computational efficiency, and memory usage.
   - Highlight how different preprocessing strategies or feature extractors influence results across datasets.
3. **Interpretation and Discussion (10%)**
   - Provide a thorough interpretation of the experimental results, identifying strengths and weaknesses of each model.
   - Critically appraise the implications of cross-dataset results (e.g., domain shift, data quality differences, representativeness).
   - Discuss observed trade-offs (e.g., accuracy vs. interpretability, complexity vs. efficiency).
   - Reflect on the limitations of the models and datasets, and propose possible improvements or directions for future research.

## 4 Report Writing (20%)

1. **Organization and Clarity (5%)**
    - Structure the report in a clear and logical manner, ensuring that each section flows naturally to the next.
    - Use straightforward language and explanations to make complex ideas accessible.
2. **Accuracy and Detail (10%)**
    - Provide precise and detailed descriptions of the models, methods, and results.
    - Ensure technical content is correct and well-supported by evidence or references.
    - Include relevant code snippets, figures, tables, and other supplementary materials that enhance understanding.
    - Ensure that all supplementary materials are clearly labeled and appropriately referenced within the report.
3. **Formatting and Word Limit (5%)**
    - Format the report professionally with consistent use of headings, fonts, and styles.
    - Ensure that all sections are written succinctly while covering all necessary details.
    - The report should be concise and should not exceed **2000 words**.

END

**DTS402TC Machine Learning**
**Group Coursework**

# Marking Criteria

| | Excellent (80-100%) | Good (60-79%) | Satisfactory (40-59%) | Poor (0-39%) |
|---|---|---|---|---|
| **1. Problem Selection & Data Collection (20%)** | | | | |
| **1.1 Problem Selection (5%)** | • Highly relevant and significant;<br>• Well-justified with clear explanation of challenges.<br>***(5 Marks)*** | • Relevant with minor issues;<br>• Adequate explanation of challenges.<br>***(4 Marks)*** | • Basic relevance;<br>• Somewhat unclear justification.<br>***(3 Mark)*** | • Not relevant;<br>• Poor or missing justification.<br>***(0-2 Marks)*** |
| **1.2 Dataset Selection (5%)** | • Meets all criteria (≥20 dims, ≥2000 data points in open-source; ≥1000 collected);<br>• Thorough justification; download address clearly provided;<br>• Strong details of collection procedure and quality assurance.<br>***(5 Marks)*** | • Mostly meets criteria with minor issues.<br>• Adequate justification.<br>• Some issues with clarity.<br>***(4 Marks)*** | • Minimum requirements met.<br>• Limited justification.<br>• Download address missing or unclear.<br>***(3 Marks)*** | • Inappropriate dataset;<br>• Poor or missing justification;<br>• Missing download address.<br>***(0-2 Marks)*** |
| **1.3 Dataset Comparison (5%)** | • Comprehensive comparison of open-source and collected data;<br>• Includes statistical summaries, visualizations, and critical discussion of representativeness.<br>***(5 Marks)*** | • Clear comparison with appropriate metrics/visuals;<br>• some discussion of differences.<br>***(4 Marks)*** | • Basic comparison;<br>• limited use of metrics / visuals;<br>• little discussion of implications.<br>***(3 Marks)*** | • Minimal or missing comparison.<br>***(0-2 Marks)*** |
| **1.4 Data Preprocessing (5%)** | • Thorough preprocessing (cleaning, normalization, encoding, missing value handling, domain alignment);<br>• Well-documented and justified.<br>***(5 Marks)*** | • Adequate preprocessing and documentation;<br>• Mostly justified.<br>***(4 Marks)*** | • Basic preprocessing with limited documentation.<br>***(3 Marks)*** | • Little or no preprocessing;<br>• Poor documentation.<br>***(0-2 Marks)*** |
| **2. Model Selection & Implementation (30%)**<br>*(Marks are based on justification and implementation, NOT on model difficulty.)* | | | | |
| **2.1 Model Selection & Construction (10%)** | • Appropriate model from a distinct category;<br>• Well-justified choice;<br>• Correct use of feature extractors for unstructured data; | • Relevant model choice with adequate justification;<br>• Implementation mostly correct with minor issues. | • Basic model implementation with limited justifications;<br>• Some core components missing; | • Inappropriate model;<br>• Missing or incorrect core components;<br>• Weak justification. |

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| | • Clearly explained and well-documented pipeline.<br>***(8-10 Marks)*** | ***(6-7 Marks)*** | • Basic explanation.<br>***(4-5 Marks)*** | ***(0-3 Marks)*** |
| **2.2 Model Training & Reporting (15%)** | • Model is thoroughly trained, validated, and tested.<br>• Includes relevant code snippets.<br>• Detailed and clear explanation of implementation in the report.<br>***(12-15 Marks)*** | • Model mostly trained, validated, and tested. Includes relevant code snippets.<br>• Clear explanation with minor issues.<br>***(9-11 Marks)*** | • Model partially trained, validated, and tested. Includes some code snippets.<br>• Basic explanation with notable issues.<br>***(6-8 Marks)*** | • Model poorly trained or not validated and tested.<br>• Incomplete or unclear code snippets.<br>• Poor explanation.<br>***(0-5 Marks)*** |
| **2.3 Code Quality (5%)** | • Code is clear, well structured, and fully documented.<br>• Consistent coding style and good programming practices throughout.<br>***(5 Marks)*** | • Code is mostly clear and structured, with minor issues in readability or document-tation.<br>• Generally consistent coding style.<br>***(4 Marks)*** | • Code shows basic structure but has limited comments or inconsistent style.<br>• Readability could be improved.<br>***(3 Marks)*** | • Code is unclear, poorly structured, lacks comments, or shows major style inconsistencies.<br>• Difficult to follow.<br>***(0-2 Marks)*** |
| **3. Result Analysis & Evaluation (30%)** | | | | |
| **3.1 Evaluation Metrics (10%)** | • Comprehensive and appropriate metrics used.<br>• Detailed and well-justified application.<br>***(8-10 Marks)*** | • Mostly appropriate metrics used.<br>• Metrics are mostly justified.<br>***(6-7 Marks)*** | • Basic metrics used with some appropriateness.<br>• Limited justification for metrics.<br>***(4-5 Marks)*** | • Inappropriate or missing metrics.<br>• Poor or no justification for metrics.<br>***(0-3 Marks)*** |
| **3.2 Comparative Analysis (10%)** | • Detailed comparison with insightful observations.<br>• Comprehensive analysis of trade-offs and conditions.<br>***(8-10 Marks)*** | • Good comparison with clear observations.<br>• Adequate analysis of trade-offs.<br>***(6-7 Marks)*** | • Basic comparison with limited observations.<br>• Incomplete analysis of trade-offs.<br>***(4-5 Marks)*** | • Poor comparison.<br>• Little to no analysis of trade-offs.<br>***(0-3 Marks)*** |
| **3.3 Interpretation and Discussion (10%)** | • Thorough analysis and insightful discussion.<br>• Clear conclusions and reflection on limitations.<br>***(8-10 Marks)*** | • Good analysis and discussion.<br>• Clear conclusions with some reflection.<br>***(6-7 Marks)*** | • Basic analysis with limited discussion.<br>• Some conclusions and limited reflection.<br>***(4-5 Marks)*** | • Poor analysis.<br>• Little to no discussion or conclusions.<br>• No reflection on limitations.<br>***(0-3 Marks)*** |
| **4. Report Writing (20%)** | | | | |
| **4.1 Organization, Clarity and** | • Clear, logical structure.<br>• Complex ideas well-explained. | • Mostly clear and organized.<br>• Most complex ideas explained. | • Adequate clarity and organization.<br>• Some complex ideas not fully explained. | • Poor clarity and organization.<br>• Complex ideas poorly explained. |

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| **Formating (10%)** | • Professional formatting with consistent headings, fonts, and styles.<br>• Report within word limit.<br>***(8-10 Marks)*** | • Good formatting with minor inconsistencies.<br>• Report within word limit.<br>***(6-7 Marks)*** | • Basic formatting with notable inconsistencies.<br>• Report exceeds or falls short of the word limit.<br>***(4-5 Marks)*** | • Poor formatting.<br>• Many sections missing or significantly over/under word limit.<br>***(0-3 Marks)*** |
| **4.2 Accuracy and Detail (10%)** | • Precise descriptions.<br>• Accurate and well-supported.<br>• All materials clearly labelled.<br>***(8-10 Marks)*** | • Accurate with minor missing details.<br>• Mostly correct and supported.<br>• Most materials labelled.<br>***(6-7 Marks)*** | • Basic descriptions with details missing.<br>• Some inaccuracies.<br>• Poorly labelled materials.<br>***(4-5 Marks)*** | • Inaccurate descriptions.<br>• Poorly supported.<br>• Missing or unclear materials.<br>***(0-3 Marks)*** |